Stowing method

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Stowing

- When coal is extracted and the void is packed with sand or other material
- Application
  - Surface subsidence is not allowed to protect surface features like Railway, Road, HV lines, Gas mains, Rivers, Water bodies, Important surface structures
  - Where the owner does not have surface rights and there is likelihood of surface subsidence
  - Cavability of roof is very poor
  - High in-situ gas content

Advantages

- No chances of accumulation of gas in goaf. Suitable for working gassy seams.
- Ventilation leakage through goaf minimum
- Front abutment pressure on workings and galleries less
- Better roof control
- Effect of surface subsidence minimised
- Upper seam workings are not affected
- Higher rate of percentage of extraction
- Possible to work below fire area or waterlogged workings
- Suitable for multi-lift workings in thick seams or depillaring by slicing method
- Depillaring of contiguous seams

Types of goaf treatment

- Hand packing
- Hydraulic stowing
  - Sand
  - Ash pond
  - Mill tailings
- Pneumatic stowing
- Mechanical packing

Difficulty in hydraulic sand stowing

- Availability of sand
- Not suitable for seams flatter than 5^0
- Not suitable for roof and floors affected by water
- Problem of increased humidity
- Pumping requirement increased
- Risk of inrush of sand-water or filling materials flowing into workings due to failure of system

Difficulty in hydraulic sand stowing

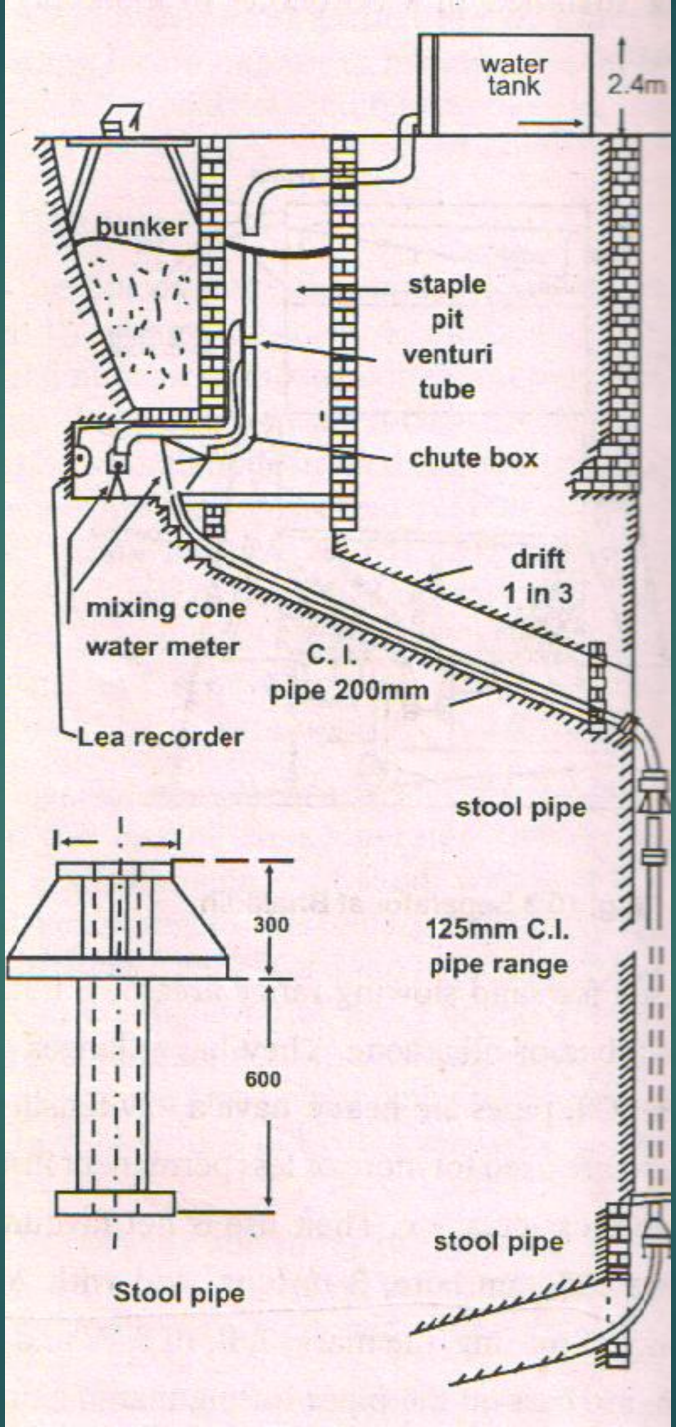
- Additional cost of sand procurement and process of sand stowing
- Require special skill for installation and maintenance
- Additional manpower requirement
- Production dependent on rate of stowing
- Blockage of pipe lines
- Cavitation in pipe line
- Wear and tear of pipe lines
- Risk of bursting of pipe line

Steps in stowing

- Gathering of sand from river bed
- Transport to mine
- Transport to bunkers
- Mixing of sand and water
- Transport of sand-water slurry
- Erection of barricade in workings
- Filling of voids
- Allowing settling of fill materials
- Shifting of pipe lines

Hydraulic profile and H:L ratio

- Pipe line should follow correct hydraulic profile
- Incorrect profile will cause cavitation full available head can not be pressured to use
- Entrapped air will cause pulsations in the system
- The length of pipe range through which sand water slurry can be transported depends mainly on the vertical head or height between mixing chamber and point of discharge
- H/L ratio of 1/7 is preferred


water tank
2.4m
bunker
staple pit
venturi tube
chute box
drift 1 in 3
mixing cone
water meter
C. I. pipe 200mm
Lea recorder
stool pipe
300
600
125mm C.I. pipe range
stool pipe
Stool pipe